

Лабораторная работа №2

Простые сети в GNS3. Анализ трафика

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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Целью данной работы является приобретение практических навыков моделирования простых компьютерных сетей в GNS3, настройки IP-адресации и анализа сетевого трафика с помощью Wireshark.

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- построить сеть с маршрутизатором VyOS;
- настроить интерфейс маршрутизатора VyOS и проверить соединение.

2. 2. Моделирование сети на базе коммутатора

2.1 Создаю топологию сети

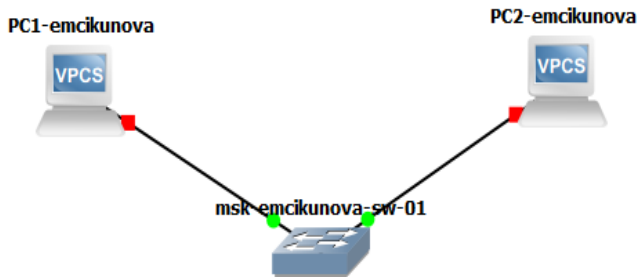
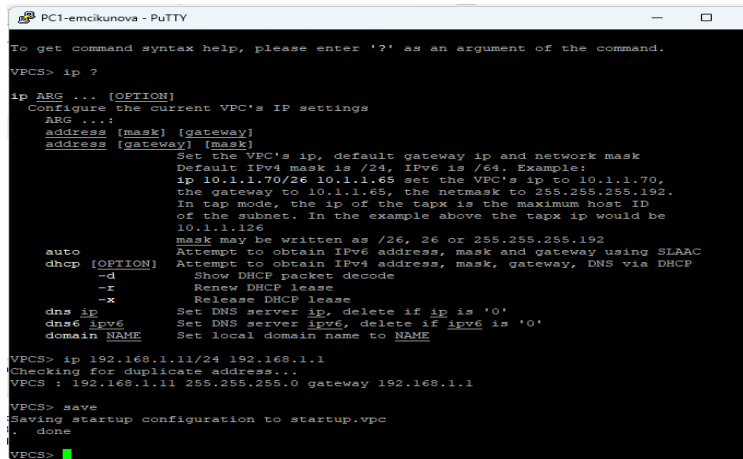


Рисунок 1: Топология сети на базе Ethernet-коммутатора

2.2 Настраиваю IP-адресацию PC1



```
PC1-emcikunova - PuTTY

To get command syntax help, please enter '?' as an argument of the command.

VPCS> ip ?

ip ARG ... [OPTION]
  Configure the current VPC's IP settings
  ARG ...:
    address [mask] [gateway]
    address [gateway] [mask]
    Set the VPC's ip, default gateway ip and network mask
    Default IPv4 mask is /24, IPv6 is /64. Example:
    ip 10.1.1.70/26 10.1.1.65 set the VPC's ip to 10.1.1.70,
    the gateway to 10.1.1.65, the netmask to 255.255.255.192.
    In tap mode, the ip of the tapx is the maximum host ID
    of the subnet. In the example above the tapx ip would be
    10.1.1.126
    mask may be written as /26, 26 or 255.255.255.192
  auto      Attempt to obtain IPv6 address, mask and gateway using SLAAC
  dhcp [OPTION]
    Attempt to obtain IPv4 address, mask, gateway, DNS via DHCP
    -d      Show DHCP packet decode
    -r      Renew DHCP lease
    -x      Release DHCP lease
  dns ip    Set DNS server ip, delete if ip is '0'
  dns6 ipv6 Set DNS server ipv6, delete if ipv6 is '0'
  domain NAME Set local domain name to NAME

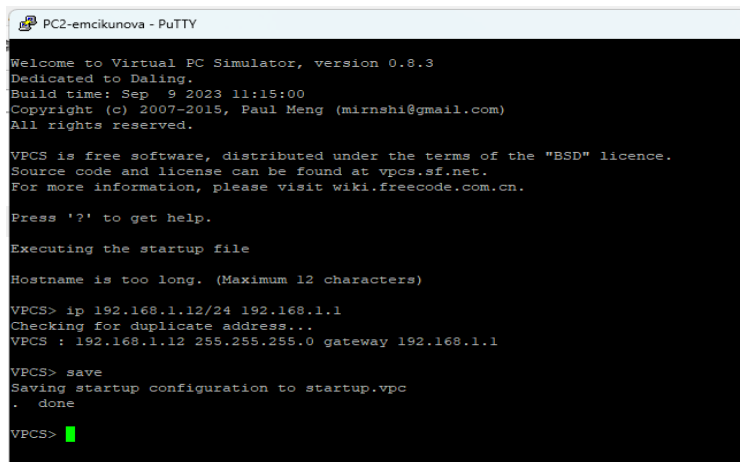
VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS>
```

Рисунок 2: Настройка IP-адреса PC1

2.3 Настраиваю IP-адресацию PC2



```
PC2-emcikunova - PuTTY

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

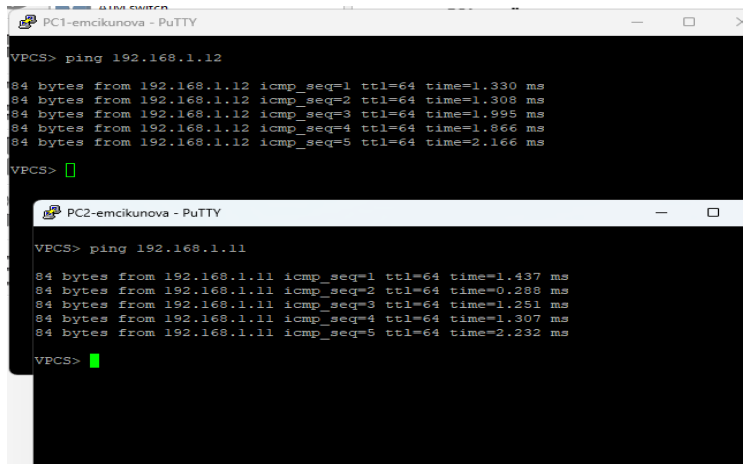
VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> █
```

Рисунок 3: Настройка IP-адреса PC2

2.4 Проверяю соединение между узлами



The image shows two overlapping PuTTY terminal windows. The top window, titled 'PC1-emcikunova - PuTTY', displays the command 'VPCS> ping 192.168.1.12' and its output: five successful ping responses from 192.168.1.12 with varying times (1.330 ms to 2.166 ms). The bottom window, titled 'PC2-emcikunova - PuTTY', displays the command 'VPCS> ping 192.168.1.11' and its output: five successful ping responses from 192.168.1.11 with varying times (1.437 ms to 2.232 ms). Both windows show a green cursor at the end of the command line.

```
PC1-emcikunova - PuTTY
VPCS> ping 192.168.1.12

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.330 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=1.308 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=1.995 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=1.866 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=2.166 ms

VPCS> █

PC2-emcikunova - PuTTY
VPCS> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.437 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.288 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.251 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.307 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=2.232 ms

VPCS> █
```

Рисунок 4: Проверка соединения между PC1 и PC2

3. 3. Анализ сетевого трафика

3.1 Анализирую ARP-трафик

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.000068	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.050754	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
4	0.051087	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
5	1.050858	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
6	1.050890	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.051859	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
8	2.052898	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)

UTC Arrival Time: Sep 19, 2026 12:15:35.908999000 UTC

Epoch Arrival Time: 1789820135.908999000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 999.771000 milliseconds]

[Time delta from previous displayed frame: 999.771000 milliseconds]

[Time since reference or first frame: 1.050858000 seconds]

Frame Number: 5

Frame Length: 64 bytes (512 bits)

Capture Length: 64 bytes (512 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:arp]

Character encoding: ASCII (0)

[Coloring Rule Name: ARP]

[Coloring Rule String: arp]

▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

- > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
- > Source: Private_66:68:01 (00:50:79:66:68:01)
 - Type: ARP (0x0806)
 - [Stream index: 3]
 - Padding: 00000000000000000000000000000000
 - Frame check sequence: 0x00000000 [unverified]
 - [PCS Status: Unverified]

▼ Address Resolution Protocol (request/gratuitous ARP)

- Hardware type: Ethernet (1)
- Protocol type: IPv4 (0x0800)
- Hardware size: 6
- Protocol size: 4
- Opcode: request (1)
 - [Is gratuitous: True]
 - Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
 - Sender IP address: 192.168.1.12
 - Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 - Target IP address: 192.168.1.12

```

0000 ff ff ff ff ff ff 00 50 79 66 68 01 08 06 00 01 .....P yfh....
0010 08 00 06 04 00 01 00 50 79 66 68 01 c0 a8 01 ec .....P yfh....
0020 ff ff ff ff ff ff c0 a8 01 0c 00 00 00 00 00 00 .....
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Рисунок 5: Анализ ARP-сообщений в Wireshark

3.2 Анализирую ICMP-трафик

No.	Time	Source	Destination	Protocol	Length	Info
4	0.051087	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
5	1.050858	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
6	1.050890	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.051859	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
8	2.052898	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
9	128.820135	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
10	128.821481	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
11	128.823979	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x677d, seq=1/256, ttl=64 (reply in 12)
12	128.825115	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x677d, seq=1/256, ttl=64 (request in 11)

Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
Destination: Private_66:68:00 (00:50:79:66:68:00)
Source: Private_66:68:01 (00:50:79:66:68:01)
Type: IPv4 (0x0800)
[Stream index: 4]
Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0x7d67 (32103)
0000 = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0x79da [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.12
Destination Address: 192.168.1.11
[Stream index: 0]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0xb88d [correct]
[Checksum Status: Good]
Identifier (BE): 26493 (0x677d)
Identifier (LE): 32103 (0x7d67)
Sequence Number (BE): 1 (0x0001)
Sequence Number (LE): 256 (0x0100)
[Response frame: 12]
Data (56 bytes)

0000 00 50 79 66 68 00 00 50 79 66 68 01 08 00 45 00 Pyfh...P yfh...E-
0010 00 54 7d 67 00 00 40 01 79 da c0 a8 01 0c c0 a8 T)g...@- y.....
0020 01 0b 05 00 b5 8d 67 7d 00 01 05 09 0a 0b 0c 0dg)
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d .. !"#\$\$% &'()*+,-
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ../012345 6789:;<=
0060 3e 3f >?

Рисунок 6: Анализ ICMP-трафика в Wireshark

3.3 Анализирую UDP-трафик

No.	Time	Source	Destination	Protocol	Length	Info
6	1.050890	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.051859	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
8	2.052808	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
9	128.820135	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
10	128.821481	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
11	128.823979	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x677d, seq=1/256, ttl=64 (reply in 12)
12	128.825115	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x677d, seq=1/256, ttl=64 (request in 11)
13	217.780118	192.168.1.12	192.168.1.11	ECHO	98	Request
14	217.780812	192.168.1.11	192.168.1.12	ECHO	98	Response

Character encoding: ASCII (0)	0000	00 50 79 66 68 00 00 50 79 66 68 01 08 00 45 00	Pyfh..P yfh...E-
[Coloring Rule Name: UDP]	0010	00 54 7d c0 00 00 40 11 79 71 c0 a8 01 0c c0 a8	T)...@ yq.....
[Coloring Rule String: udp]	0020	01 05 0a 24 00 07 00 40 d6 50 00 50 79 66 68 01	..\$....@ .P Pyfh.
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (	0030	0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d	
> Destination: Private_66:68:00 (00:50:79:66:68:00)	0040	1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d	.. !"#\$\$% &'()*+,-
> Source: Private_66:68:01 (00:50:79:66:68:01)	0050	2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d	./012345 6789:;<=
Type: IPv4 (0x0800)	0060	3e 3f	>?
[Stream index: 4]			
▼ Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11			
0100 = Version: 4			
.... 0101 = Header Length: 20 bytes (5)			
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)			
Total Length: 84			
Identification: 0x7dc0 (32192)			
> 000. = Flags: 0x0			
...0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 64			
Protocol: UDP (17)			
Header Checksum: 0x7971 [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 192.168.1.12			
Destination Address: 192.168.1.11			
[Stream index: 0]			
▼ User Datagram Protocol, Src Port: 2596, Dst Port: 7			
Source Port: 2596			
Destination Port: 7			
Length: 64			
Checksum: 0xd650 [unverified]			
[Checksum Status: Unverified]			
[Stream index: 0]			
[Stream Packet Number: 1]			
> [Timestamps]			
UDP payload (56 bytes)			
> Echo			

Рисунок 7: Анализ UDP-трафика в Wireshark

3.4 Анализирую TCP-трафик

16 288.061933 Private_66:68:00 Private_66:68:01 ARP 64 192.168.1.11 is at 00:50:79:66:68:00

17 288.062998 192.168.1.12 192.168.1.11 TCP 74 18300 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1789820422 TSecr=0 WS=2

18 288.065950 192.168.1.11 192.168.1.12 TCP 54 7 → 18300 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0

19 288.067648 192.168.1.12 192.168.1.11 TCP 66 18300 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1789820422 TSecr=0

20 288.069517 192.168.1.12 192.168.1.11 ECHO 122 Request

21 288.071298 192.168.1.11 192.168.1.12 TCP 54 7 → 18300 [ACK] Seq=1 Ack=57 Win=2920 Len=0

22 288.074998 192.168.1.12 192.168.1.11 TCP 66 18300 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1789820422 TSecr=0

23 288.075539 192.168.1.11 192.168.1.12 TCP 54 7 → 18300 [ACK] Seq=1 Ack=58 Win=2920 Len=0

24 288.075955 192.168.1.11 192.168.1.12 TCP 54 7 → 18300 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0

25 288.077960 192.168.1.12 192.168.1.11 TCP 66 18300 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1789820422 TSecr=0

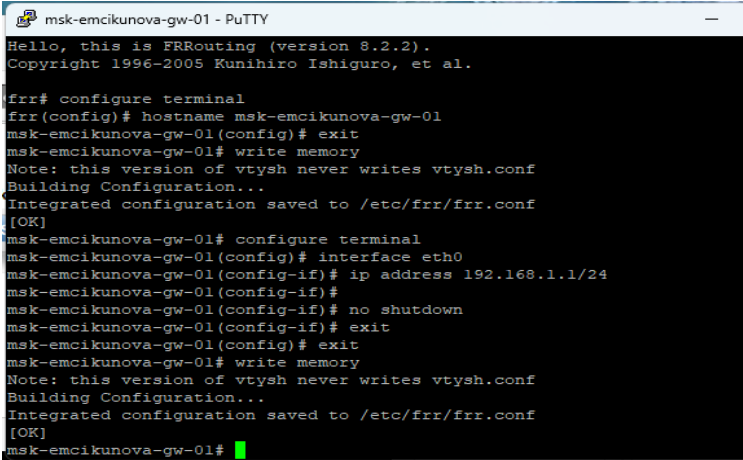
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x7955 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.12
Destination Address: 192.168.1.11
[Stream index: 0]
Transmission Control Protocol, Src Port: 18300, Dst Port: 7, Seq: 1, Ack: 1, Len: 0
Source Port: 18300
Destination Port: 7
[Stream index: 0]
[Stream Packet Number: 3]
Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1405153622
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1833534204
1000 = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 2920
[Calculated window size: 2920]
[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0xB258 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
Client Contiguous Streams: 1
Server Contiguous Streams: 1

0000 00 50 79 66 68 00 00 50 79 66 68 01 00 00 45 00 Pyfh..P yfh...E..
0010 00 34 7e 07 00 00 40 06 79 55 c0 a8 01 0c c0 a8 4a...@: yU.....
0020 01 0b 47 7c 00 07 53 c0 f1 56 6d 49 82 fc 80 10 ..G]..S..VmI....
0030 0b 68 82 58 00 00 01 01 08 0a 6a ae 7e 06 00 00 ..hX....-j.....
0040 00 00

Рисунок 8: Анализ TCP-трафика в Wireshark

4. 4. Моделирование сети с маршрутизатором FRR

4.1 Настраиваю маршрутизатор FRR



```
msk-emcikunova-gw-01 - PuTTY
Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-emcikunova-gw-01
msk-emcikunova-gw-01(config)# exit
msk-emcikunova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-emcikunova-gw-01# configure terminal
msk-emcikunova-gw-01(config)# interface eth0
msk-emcikunova-gw-01(config-if)# ip address 192.168.1.1/24
msk-emcikunova-gw-01(config-if)#
msk-emcikunova-gw-01(config-if)# no shutdown
msk-emcikunova-gw-01(config-if)# exit
msk-emcikunova-gw-01(config)# exit
msk-emcikunova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-emcikunova-gw-01#
```

Рисунок 9: Настройка маршрутизатора FRR

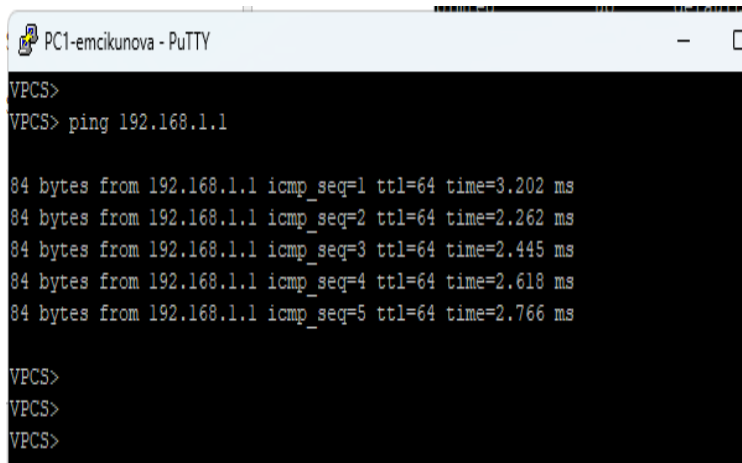
4.2 Проверяю конфигурацию FRR

```
mik-emcikunova-gw-01#  
mik-emcikunova-gw-01# show interface brief  
Interface      Status  VRF      Addresses  
-----  
eth0            up      default  192.168.1.1/24  
eth1            down    default  
eth2            down    default  
eth3            down    default  
eth4            down    default  
eth5            down    default  
eth6            down    default  
eth7            down    default  
lo              up      default  
pimreg          up      default  
  
mik-emcikunova-gw-01#
```

Рисунок 10: Проверка интерфейсов маршрутизатора FRR

4.3 Проверяю соединение с FRR

С узла **PC1** выполняю **ping 192.168.1.1**. Маршрутизатор успешно отвечает на ICMP-запросы.



```
PC1-emcikunova - PuTTY
VPCS>
VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=3.202 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=2.262 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=2.445 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=2.618 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=2.766 ms

VPCS>
VPCS>
VPCS>
```

4.4 Анализирую трафик FRR

No.	Time	Source	Destination	Protocol	Length	Info
→	17 196.329890	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x637f, seq=1/256, ttl=64 (reply in 18)
←	18 196.332605	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x637f, seq=1/256, ttl=64 (request in 17)
	19 197.334213	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x647f, seq=2/512, ttl=64 (reply in 20)
	20 197.335685	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x647f, seq=2/512, ttl=64 (request in 19)
	21 198.337422	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x657f, seq=3/768, ttl=64 (reply in 22)
	22 198.339032	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x657f, seq=3/768, ttl=64 (request in 21)
	23 199.340439	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x667f, seq=4/1024, ttl=64 (reply in 24)
	24 199.342201	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x667f, seq=4/1024, ttl=64 (request in 23)
	25 200.343956	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x677f, seq=5/1280, ttl=64 (reply in 26)
	26 200.345675	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x677f, seq=5/1280, ttl=64 (request in 25)

▼ Ethernet II, Src: 0c:0b:52:3f:00:00 (0c:0b:52:3f:00:00), Dst: Private_66:68:00 > Destination: Private_66:68:00 (00:50:79:66:68:00) > Source: 0c:0b:52:3f:00:00 (0c:0b:52:3f:00:00) > Type: IPv4 (0x0800) [Stream index: 4] ▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.10 0100 = Version: 4 0101 = Header Length: 20 bytes (5) > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 84 Identification: 0x6fe5 (28645) > 000. = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 64 Protocol: ICMP (1) Header Checksum: 0x0768 [validation disabled] [Header checksum status: Unverified] Source Address: 192.168.1.1 Destination Address: 192.168.1.10 [Stream index: 0] ▼ Internet Control Message Protocol Type: Echo (ping) reply (0) Code: 0 Checksum: 0xc48b [correct] [Checksum Status: Good] Identifier (BE): 25471 (0x637f) Identifier (LE): 32611 (0x7f63) Sequence Number (BE): 1 (0x0001) Sequence Number (LE): 256 (0x0100) [Request frame: 17] [Response time: 2,715 ms] > Data (56 bytes) 0000 00 50 79 66 68 00 0c 0b 52 3f 00 00 08 00 45 00 -Pyfh... R?...E- 0010 00 54 6f e5 00 00 40 01 87 68 c0 a8 01 01 c0 a8 -To...@...h...- 0020 01 0a 00 00 c4 8b 63 7f 00 01 08 09 0a 0b 0c 0d -...c...- 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d -...- 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d -...!"#\$% &'()*+,- 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d -./012345 6789;:<= 0060 3e 3f ->
--

Рисунок 12: Анализ трафика между PC1 и FRR

5. 5. Моделирование сети с маршрутизатором VyOS

5.1 Создаю топологию с VyOS

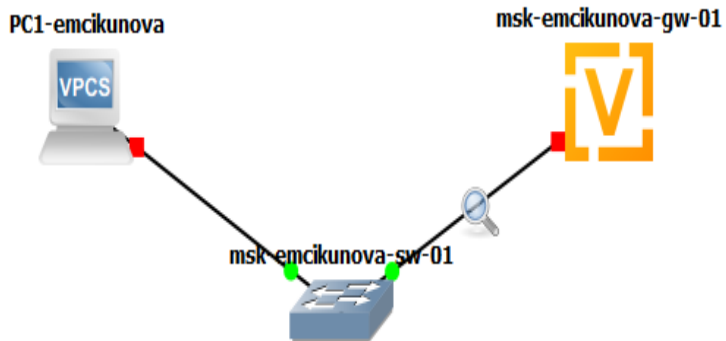
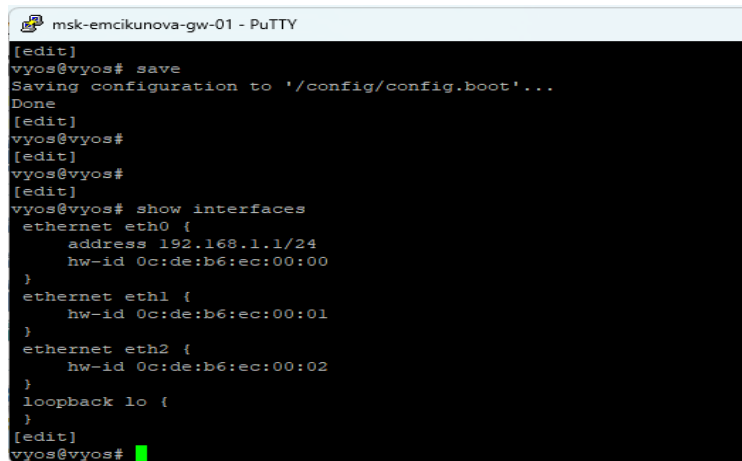


Рисунок 13: Топология сети на базе маршрутизатора VyOS

5.2 Настраиваю маршрутизатор VyOS

Для интерфейса **eth0** отключаю получение адреса по DHCP и задаю статический IP-адрес **192.168.1.1/24**. Применяю и сохраняю конфигурацию.

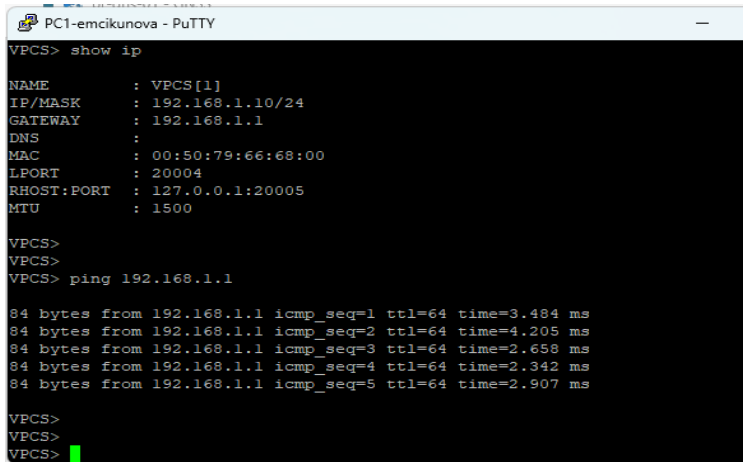
5.3 Проверяю интерфейсы VyOS



```
msk-emcikunova-gw-01 - PuTTY
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
[edit]
vyos@vyos#
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:de:b6:ec:00:00
  }
  ethernet eth1 {
    hw-id 0c:de:b6:ec:00:01
  }
  ethernet eth2 {
    hw-id 0c:de:b6:ec:00:02
  }
  loopback lo {
  }
[edit]
vyos@vyos#
```

Рисунок 14: Проверка интерфейсов маршрутизатора VyOS

5.4 Проверяю соединение с VyOS



```
PC1-emcikunova - PuTTY
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT : 127.0.0.1:20005
MTU        : 1500

VPCS>
VPCS>
VPCS> ping 192.168.1.1

64 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=3.484 ms
64 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=4.205 ms
64 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=2.658 ms
64 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=2.342 ms
64 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=2.907 ms

VPCS>
VPCS>
VPCS> █
```

Рисунок 15: Проверка соединения с маршрутизатором VyOS

5.5 Анализирую трафик VyOS

No.	Time	Source	Destination	Protocol	Length	Info
22	204.712744	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
23	204.716558	0c:ide:b6:ec:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at 0c:ide:b6:ec:00:00
24	204.718204	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xe180, seq=1/256, ttl=64 (reply in 25)
25	204.720700	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe180, seq=1/256, ttl=64 (request in 24)
26	205.722606	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xe200, seq=2/512, ttl=64 (reply in 27)
27	205.725427	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe200, seq=2/512, ttl=64 (request in 26)
28	206.727967	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xe300, seq=3/768, ttl=64 (reply in 29)
29	206.729697	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe300, seq=3/768, ttl=64 (request in 28)
30	207.732316	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xe400, seq=4/1024, ttl=64 (reply in 31)
31	207.733702	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe400, seq=4/1024, ttl=64 (request in 30)

[Coloring Rule String: icmp || icmpv6]

▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:ide:b6:ec:00:00

> Destination: 0c:ide:b6:ec:00:00 (0c:ide:b6:ec:00:00)

> Source: Private_66:68:00 (00:50:79:66:68:00)

Type: IPv4 (0x0800)

[Stream index: 5]

▼ Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1

0100 = Version: 4

... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x00e2 (32994)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x766b [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.10

Destination Address: 192.168.1.1

[Stream index: 1]

▼ Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0x3d89 [correct]

[Checksum Status: Good]

Identifier (BE): 57984 (0xe280)

Identifier (LE): 32994 (0x00e2)

Sequence Number (BE): 2 (0x0002)

Sequence Number (LE): 512 (0x0200)

[Response frame: 27]

> Data (56 bytes)

0000 0c de b6 ec 00 00 00 50 79 66 68 00 00 00 45 00P yfh...E:

0010 00 54 80 e2 00 00 40 01 76 6b c0 a8 01 0a c0 a8 ...T....@ vk.....

0020 01 01 08 00 3d 89 e2 80 00 02 08 09 0a 0b 0c 0d=.....

0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d

0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ... !"#\$\$% &'()*+,-

0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d/012345 6789;:<=

0060 3e 3f>?

Рисунок 16: Анализ ARP- и ICMP-трафика между PC1 и VyOS

6. 6. Вывод

6.1 Результаты лабораторной работы

В ходе лабораторной работы были построены простые сети в GNS3 на базе Ethernet-коммутатора и маршрутизаторов FRR и VyOS.

Была настроена IP-адресация и проверена связность между устройствами. С помощью Wireshark был исследован сетевой обмен по протоколам ARP, ICMP, UDP и TCP.

Настроенные сети функционировали корректно, а передаваемые пакеты успешно достигали адресатов.